LABORATORY RESULT REPORT

Lab 02: Reservoir Stage Kinematics & Logistic Modeling

Student Name:	Miranda	**Course:**	Numerical Solution to Civil Engineering Problems
Section:	BSCE-3D	**Dataset:**	Data-01.xlsx

1. EXECUTIVE SUMMARY & KEY PERFORMANCE INDICATORS

This laboratory report presents a comprehensive numerical analysis of reservoir stage kinematics and non-linear volumetric response using measured stage-time telemetry dataset `Data-01.xlsx`. High-order numerical differentiation, non-linear regression via a four-parameter logistic model, and adaptive numerical integration were executed to characterize inflow behavior and cross-validate cumulative storage capacity.

KEY PERFORMANCE INDICATOR (KPI)	QUANTITATIVE VALUE	ENGINEERING SIGNIFICANCE / DESCRIPTION
Analysis Duration (t)	`71.75 hrs`	Total monitoring interval (t_0 = 0.00 h to t_f = 71.75 h)
Peak Inflow Velocity (dh/dt_{max})	`0.9600 m/hr`	Maximum instantaneous rate of stage change, recorded at t = 29.25 h
Coefficient of Determination (R^2)	`0.979149`	Goodness-of-fit metric for the non-linear logistic regression
Standard Error of Estimate (s)	`0.386445 m`	Root-mean-square deviation of measured stage relative to the model
Cumulative Stage-Time Integral	`1260.923879 m·hr`	Total area under the stage curve evaluated via adaptive quadrature

2. NUMERICAL DIFFERENTIATION & KINEMATIC INSIGHTS

Finite-difference numerical schemes were implemented to evaluate the first derivative (velocity, dh/dt) and second derivative (acceleration, d^2h/dt^2) of stage elevation with respect to time:

- **Velocity Dynamics (dh/dt):** The peak inflow response occurs at t = 29.25 hours, yielding a maximum stage velocity of **0.9600 m/hr**.
- **Acceleration Transition (d^2h/dt^2):** The second derivative curve exhibits a distinct zero-crossing point (d^2h/dt^2 = 0) near t = 29.25 hours. This critical inflection point demarcates the boundary where inflow velocity transitions from positive acceleration to deceleration.

3. NON-LINEAR MODEL FITTING (4-PARAMETER LOGISTIC FIT)

The non-linear reservoir stage elevation $h(t)$ is modeled using a four-parameter logistic formulation:

$$h(t) = c + {}^{a}/_{(1 + e^{-k(t - t_0)})}$$

PARAMETER	PHYSICAL INTERPRETATION	ESTIMATE VALUE	STANDARD ERROR
Scale Parameter (a)	Total stage capacity range	5.513894 m	±0.049942
Growth Rate (k)	Steepness factor of stage rise	0.855571 hr^{-1}	±0.062457
Inflection Point (t_0)	Midpoint time of peak inflection	30.157709 hr	±0.097845
Baseline Offset (c)	Initial baseline stage elevation	14.377539 m	±0.037905

4. STATISTICAL PARAMETER DIAGNOSTICS

Two-tailed Student's t-tests were conducted to evaluate the statistical significance of each parameter estimate within the non-linear regression framework.

PARAMETER NAME	T-STATISTIC	P-VALUE	STATISTICAL STATUS (Α = 0.05)
Scale Parameter (a)	110.4058	< 0.0001	SIGNIFICANT
Growth Rate (k)	13.6987	< 0.0001	SIGNIFICANT
Inflection Point (t_0)	308.2197	< 0.0001	SIGNIFICANT
Baseline Offset (c)	379.3039	< 0.0001	SIGNIFICANT

Regression Diagnostic Summary:
• Sum of Squared Errors (SSE): 42.412537 | • Total Sum of Squares (SST): 2034.077941 | • Degrees of Freedom (dof): 284

5. VOLUMETRIC INTEGRATION & ALGORITHM CROSS-VALIDATION

To evaluate cumulative exposure area over the duration $t \in [0, 71.75]$, discrete trapezoidal numerical integration was rigorously cross-validated against continuous adaptive Gauss-Kronrod quadrature ($\int h(t)\, dt$).

INTEGRATION ALGORITHM	CALCULATED VALUE	ABSOLUTE ERROR (M·HR)	RELATIVE ERROR
Adaptive Quadrature (`scipy.integrate.quad`)	1260.923879 m·hr	Reference Baseline	Base Method
Trapezoidal Integration (`numpy.trapezoid`)	1260.923879 m·hr	2.273737×10^{-13}	0.000000%

6. ENGINEERING CONCLUSION

- **Model Fidelity:** The four-parameter logistic formulation proves highly robust for reservoir stage estimation, accounting for 97.91% of total dataset variance ($R^2 = 0.979149$) with all optimized parameters exhibiting high statistical significance ($p < 0.0001$).
- **Kinematic Alignment:** Finite-difference derivatives confirm that peak inflow velocity (0.9600 m/hr at $t = 29.25$ h) corresponds precisely to acceleration zero-crossing, aligning closely with the fitted logistic midpoint ($t_0 = 30.16$ h).
- **Numerical Precision:** Volumetric cross-validation demonstrates exact numerical concordance between continuous quadrature and discrete trapezoidal methods (0.0000% relative error), validating the stability and precision of the discretization scheme for civil engineering hydraulic routing applications.